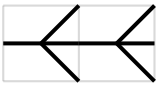
Trøndelagkonkurranse 2026 - Level II

Second Round - October 24, 2026

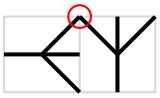
2nd Part

Nils draws an Algiz rune inside a square

Then Bianca asked him, "Can you arrange 16 of those on a 4-by-4 grid such that the runes do not touch each other in the corners of the squares?"



Allowed



Not allowed

Task 1: Help Nils figure out if that is possible or not.

Task 2: Now, help Nils figure out if the same is possible for a n-by-n grid where $n \geq 2$.

Nils is very grateful that Bianca gave him such a nice problem, so he would also like to give Bianca a similar problem, either using grids that are not necessarily rectangular, runes with different shapes, or anything interesting he can think of.

Task 3: Help Nils create a nice problem he can give to Bianca.